CEFTAZIDIME

Antipseudomonal

Cross-allergy

CHEMICAL NAME (6R,7R)-7-[[(2Z)-2-(2-amino-1,3-thiazol-4-yl)-2-(2-carboxypropan-2-yloxyimino)acetyl]amino]-8-oxo-3-(pyridin-1-ium-1-ylmethyl)-5-thia-1-azabicyclo[4.2.0]oct-2-ene-2-carboxylate (pentahydrate)

CLASS Third-generation cephalosporin with antipseudomonal activity

TRADE NAMES Fortum · Fortaz · Tazicef · Zavicefta (with avibactam)

THERAPEUTIC USES / INDICATIONS

- *Pseudomonas aeruginosa* infections: hospital-acquired pneumonia, cystic fibrosis exacerbations.
- Febrile neutropenia (empirical).
- Complicated UTI; meningitis due to Gram-negatives.
- Melioidosis.

ADMINISTRATION

- **IV** injection over 3–5 min or infusion over 15–30 min; **IM**.
- Reconstitution releases CO_2 — follow vial instructions.
- Extended infusions may improve efficacy.

DOSAGES

- **Adult:** 1–2 g every 8 h (max 9 g/day in cystic fibrosis).
- **Children:** 25–50 mg/kg every 8 h; cystic fibrosis up to 150 mg/kg/day.
- **Renal impairment:** CrCl 31–50: 1 g every 12 h; 16–30: 1 g every 24 h; lower below.

PHARMACOKINETICS

- **Onset:** immediate (IV). Good CSF penetration with inflamed meninges.
- Not metabolised.
- Excreted unchanged in urine. **t½:** ~2 h (prolonged in renal failure).

PHARMACODYNAMICS

- Inhibits **cell-wall synthesis** → bactericidal.
- Strong Gram-negative activity including *Pseudomonas*; weak against Gram-positives and anaerobes.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea.
- Phlebitis, injection pain.
- Rash, raised eosinophils.

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis.
- Neurotoxicity (encephalopathy, seizures) in renal failure without dose reduction.
- *C. difficile* colitis.
- Thrombocytosis, leucopenia.

CONTRAINDICATIONS & CAUTIONS

- Cephalosporin hypersensitivity; severe penicillin allergy.
- **Caution:** renal impairment (adjust dose).

KEY INTERACTIONS

- **Aminoglycosides, loop diuretics:** ↑ nephrotoxicity.
- **Chloramphenicol:** antagonism.

NURSING MANAGEMENT

- Check allergy history and renal function before dosing.
- Monitor neurological status in renal impairment.
- Monitor temperature, cultures, CBC.
- Monitor IV site.
- Watch for diarrhoea.

PATIENT EDUCATION

- Report confusion, twitching, rash or severe diarrhoea.

KEY POINTS

- 3rd-gen cephalosporin active against Pseudomonas.
- Adjust dose in renal impairment (neurotoxicity).
- Weak Gram-positive cover.
- Febrile neutropenia, cystic fibrosis.

SOURCES CHECKED **[141]** Ceftazidime 1g Powder for Solution for Injection or Infusion (UK SmPC) **[142]** Ceftazidime, IM, IV (US label) **[143]** Ceftazidime, IM, IV (US label, WG Critical Care)

Full references p. 432–457